

Zhongyu Fan

Sc.M. Student in Biostatistics
Department of Biostatistics, Brown University
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SELF SUMMARY

Biostatistics master's student with experience in R/Shiny tool development, statistical modeling, simulation, biomedical machine learning, and clinical evidence review. Interested in applying reproducible computing and AI-assisted validation tools to clinical data programming, quality control, and regulatory drug development.

EDUCATION

Expected 06/2027	Sc.M. in Biostatistics Brown University, School of Public Health Thesis Advisor: Dr. Ying Ma
09/2021–06/2025	B.S. (Honors) in Mathematics with Applied Mathematics University of Nottingham Ningbo, School of Mathematical Sciences Thesis Advisors: Dr. Elio Eduardo Espejo Arenas and Dr. Mainul Haque

PROJECT EXPERIENCE (SELECTED)

Interactive R/Shiny ABM–SEIR Infectious Disease Simulation Platform	Providence, RI, USA
11/2025–12/2025	R/Shiny App

- Developed an interactive R/Shiny application integrating agent-based SEIR simulation, literature-driven epidemiological parameter presets, and user-defined intervention scenarios.
- Implemented Monte Carlo simulation modules to generate synthetic outbreak trajectories across seven intervention scenarios, summarizing epidemic curves, peak prevalence, peak timing, and 95% confidence intervals.
- Built regression, diagnostic, visualization, and CSV export modules, including interaction terms, coefficient plots, correlation heatmaps, and spatial animations to support reproducible scenario testing.
- Designed the application workflow to support user-friendly parameter input, automated output generation, and transparent review of simulation assumptions and results.

Relational Database Design and Tableau Business Analytics Project	Ningbo, China
02/2024–05/2024	Team Leader

- Designed and implemented a relational database containing 3,000+ customer and sales records to support structured querying, analysis, and visualization.
- Built 3 interactive Tableau dashboards and 1 data story, supplemented by R-based visualizations, to translate complex data into user-facing analytical insights.

RESEARCH EXPERIENCE (SELECTED)

The Ma Lab, Department of Biostatistics, Brown University	Providence, RI, USA
10/2025–Present	Research Assistant, Advised by Dr. Ying Ma and Dr. Boran Gao

- Develop graph-regularized sparse matrix factorization methods for GWAS summary statistics incorporating biological annotations.
- Implement reproducible R/Rcpp workflows for algorithm development, simulation studies, real-data analysis, enrichment analysis, factor validation, and visualization of biomedical patterns.
- Evaluate model outputs through hyperparameter comparison, factor interpretation, biological enrichment, and performance summaries to support robust biomedical data analysis.

Global Mental Health and Flourishing Lab, Brown University	Providence, RI, USA
11/2025–Present	Research Assistant, Advised by Dr. Shufang Sun

- Conduct longitudinal statistical analyses on a youth–caregiver cohort study following approximately 2,200 Ukrainian youth and 1,900 parents/caregivers to examine war exposure, mental health outcomes, and risk behaviors.
- Apply logistic GEE models to evaluate associations between war-related exposures and youth mental health outcomes, including depression, anxiety, PTSD, loneliness, and suicidality.
- Explore multivariable exposure modeling and LASSO-based variable selection to identify war-related risk factors associated with high-risk youth mental health profiles.

PROFESSIONAL EXPERIENCE

IQVIA plc	Shanghai, China
11/2024–01/2025	Analyst Intern, Management Consulting Department

- Conducted oncology-focused clinical evidence research on immunotherapy assets, synthesizing clinical trial publications, treatment guidelines, and competitive intelligence for multinational pharmaceutical clients.
- Reviewed 20+ clinical trial papers and guidelines; extracted structured information on study design, patient population, endpoints, efficacy, safety, and treatment positioning.

- Conducted 10+ KOL/external expert interviews and analyzed 200+ clinical stakeholder survey responses to identify unmet clinical needs and treatment decision patterns.
- Prepared structured evidence summaries, analytical tables, figures, and client-facing deliverables with attention to consistency, traceability, and quality control of analytical outputs.

HALMA plc

Shanghai, China

07/2024–10/2024 Business Intelligence and M&A Research Intern, Healthcare Sector

- Extracted and processed disease-level data for 300 conditions from the Global Burden of Disease database; built regression-based analyses using incidence and patient age to identify 105 aging-related diseases across 35 clinical categories.
- Reviewed 50+ medical literature sources and clinical guidelines to evaluate disease burden, diagnostic pathways, monitoring needs, treatment options, and unmet clinical needs for aging-related conditions.
- Screened 20+ medical device companies through the National Medical Products Administration medical device registration system, integrating regulatory, product, and market information into structured opportunity assessments.

Beijing AIMEDICINE Co., Ltd. × NYU Shanghai

Shanghai, China

07/2023–08/2023 Data Analyst

- Designed survey questionnaires for elderly residents, healthcare professionals, database maintenance personnel, and nursing home management teams in a digital health research project.
- Conducted field data collection across 73 nursing homes; established a structured database and synthesized findings for a white paper on elderly care digitalization.

SELECTED SCHOLARLY WORK

Under Review	Fan, Z., & Skeen, S. (2026). <i>Direct War Exposure, Intergenerational Risk Factors, and Youth Mental Health: A School-Based Cohort Study Three Years After the 2022 Russo-Ukrainian War</i> . Symposium proposal submitted to the 8th International Conference of the International Association for Youth Mental Health (IAYMH 2026), Dublin, Ireland.
Work in Progress	Fan, Z. (2026). GEAR: A New Regularized Matrix Factorization for Meaningful Biological Interpretation in GWAS Summary Statistics. <i>Algorithm Completed; Data analysis in progress</i> .

HONORS & AWARDS

2025–2026	Todd '87 and Christine Fisher P'17 Graduate Fellowship, Brown University	\$10,000
2024	Global Engagement Office Summer Exchange Tuition Waiver Scholarship, University of Nottingham Ningbo	€1,800

ACADEMIC SERVICES

06/2024–07/2024	Teaching Assistant, MATH3048: Mathematical Medicine and Biology, University of Nottingham Ningbo
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SKILLS

Programming & Tool Development: R, Shiny, R Markdown/Quarto, Python, SQL, C++, Rcpp, Git, Bash/Zsh, L^AT_EX, Markdown

R Skills/Packages: Shiny, tidyverse, data.table, ggplot2, kableExtra, lme4, geepack, brms, glmnet

Statistical Modeling & Simulation: Logistic regression, GLM, GEE, mixed-effects models, longitudinal and multilevel data analysis, Monte Carlo simulation, LASSO/ridge regularization, matrix factorization, graph regularization.

Data Validation & Visualization: Quality control of analytical tables and figures, reproducible reporting, diagnostic visualization, CSV export, Tableau dashboards, Excel Pivot Tables, PowerPoint.

Clinical Data & Regulatory Awareness: Clinical trial literature review, treatment guideline review, oncology and immunotherapy evidence synthesis, familiarity with TFL concepts, awareness of CDISC, SDTM, and ADaM concepts, NMPA medical device registration system.

Communication: Mandarin Chinese (Native), English (Proficient).

May 13, 2026